Rutvi Khamar

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Summary

I'm particularly interested in AI interpretability, safety, and evaluation. I want to work on making models more transparent, reliable, and aligned, contributing to research that shapes how we build trustworthy AI systems.

Education

Florida Institute of Technology Aug 2023 - May 2025
M.S. in Computer Science (GPA: 3.93 / 4.00) *Melbourne, Florida*
Relevant Coursework: Applied Discrete Mathematics, Machine Learning (coded algorithms from scratch), Big Data (Spark/Hadoop/AWS), AI, ML engineering, Research (LLMs in Healthcare), Thesis

Adani University Aug 2019 - May 2023
B.Tech in Information and Communication Technology (GPA: 8.93 / 10.00) *Ahmedabad, Gujarat*
Relevant Coursework: Data Mining & BI, Linear Algebra, Probability & Statistics, DSA, Python for Data Science

Technical Skills

Programming: Python, SQL, Java, DSA
ML frameworks: Numpy, Pandas, PyTorch, Tensorflow, Transformers, HuggingFace, TransformerLens, Dask, LangChain, LangGraph, MLflow, Weights & Biases, scikitlearn, matplotlib, plotly, Streamlit
Research Interest and skills: Mechanistic Interpretability, Transformer Architecture, Sparse Autoencoders, Representation Learning, LLM Evaluations, Statistical Analysis, Experimental Design

Research Work

Early breast cancer detection with 3D ultrasound using NMF (Thesis) | *Advisor: Dr. Mitra*

- Published and presented this research work at the SPIE Medical Imaging Conference 2025, detailing a novel ML approach for early-stage breast cancer detection.
- Developed and optimized multiple NMF (Non-negative Matrix Factorization) variants with custom regularization to extract diagnostic features from DCE-US data, distinguishing between benign and malignant lesions.
- Achieved 75.5% sensitivity and 78.3% specificity using only pre-injection data, matching traditional contrast-enhanced performance while eliminating patient risk and streamlining clinical workflows.

Experience

Steppingblocks | *AI/ML Engineer - Melbourne, FL* Aug 2025 – Present

- Designed and implemented several product features along with their evaluation frameworks for production ML models (classification, record linkage), establishing accuracy, precision/recall metrics, and automated quality monitoring pipelines
- Built embedding-based representation learning pipelines with RAG, vector search, and cross-encoder re-ranking for data consistency, across skills and job titles at scale.
- Owned end-to-end ML lifecycle, including dataset curation, evaluation metrics, monitoring, retraining, and explainability.
- Collaborated cross-functionally to translate ambiguous product requirements into scalable ML systems with measurable impact.

L3Harris Institute for Assured Information | *Research Assistant - Melbourne, FL* May 2024 – July 2024

- Conducted systematic research comparing multiple entity extraction approaches (SpaCy+ClausIE, Stanza-CoreNLP, fine-tuned LLaMA2, T5-based transformers) to identify optimal methods for converting unstructured text into structured knowledge graphs, evaluating trade-offs between accuracy, computational cost, and scalability.
- Designed and executed experiments fine-tuning LLaMA2 for structured query generation with limited compute resources, implementing prompt engineering and parameter-efficient fine-tuning strategies to enable text-to-Cypher translation for graph database querying.
- Built end-to-end GraphRAG research prototype integrating LLMs, Neo4j graph databases, to explore how graph-structured knowledge enhances LLM reasoning over complex relational data.
- Documented comparative analysis across all approaches, evaluating each method's strengths and failure modes to guide future research directions.

Projects

High School Helper – AI Agent | *LangGraph, Groq Cloud, OpenAI OSS-120B, SymPy, Streamlit*

- Designed and implemented a multimodal foundation-model-based agent integrating vision and language processing to solve complex math and physics problems.
- Implemented a ReAct multi-agent architecture combining chain-of-thought reasoning with symbolic computation via SymPy to eliminate LLM hallucinations and ensure precise numerical results.

FlightMate | *OpenAI API, LangChain, NetworkX, Prompt Engineering*

- Developed an AI agent integrating retrieval-augmented generation (RAG) with graph-based reasoning and live data integration for structured information understanding.
- Implemented a multi-tool ReAct architecture for dynamic reasoning and visualization of flight network insights.

Embedding visualization tool | *Huggingface, Embedding models, SentenceTransformer, scikit-learn tSNE, Streamlit*

- Built an interactive tool to explore how models represent semantic relationships geometrically, using cosine similarity heatmaps, hierarchical clustering, and dimensionality reduction (t-SNE) to visualize embedding spaces.
- Implemented multiple normalization techniques (z-score, min-max, rank-based) to enable fair comparison across embedding models with different calibration scales.

Fuzzy Inference Systems for Enhanced Interpretability in Machine Learning | *Python, self-coded ML algorithms*

- Optimized fuzzy inference systems using genetic algorithms to enhance model interpretability while maintaining high predictive accuracy.
- Improved transparency of machine learning models by optimizing fuzzy membership functions for explainable, high-performing predictions.